

Eonyong Han

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Research Interests

Multi-omics data integration; computational cancer genomics; machine learning for biomedical data; transcriptomics and pathway analysis; immune-response modeling in infectious disease (COVID-19). Emphasis on translating high-throughput sequencing data into clinically interpretable insights.

Education

Ph.D. in Computer Science and Engineering 20XX – 2026 (expected)
Kyungpook National University, Daegu, Republic of Korea

- Dissertation: *[Working title – e.g., “Multi-omics integration methods for cancer subtyping and disease-progression modeling”]*
- Advisor: Prof. Inuk Jung

[B.S. / M.S.] in [Field] 20XX – 20XX
[University], [City], Republic of Korea

Publications

[†]Co-first author. Author's name in **bold**. Full metadata in `references.bib`.

Peer-reviewed journal articles

1. **Han, E.**, Youn, S., Kwon, K.T., Kim, S.C., Jo, H.-Y., & Jung, I. (2024). Disease progression associated cytokines in COVID-19 patients with deteriorating and recovering health conditions. *Scientific Reports*, 14, 24712. doi:10.1038/s41598-024-75924-x
2. **Han, E.[†]**, Kwon, H.[†], & Jung, I. (2025). A review on multi-omics integration for aiding study design of large scale TCGA cancer datasets. *BMC Genomics*, 26, 769. doi:10.1186/s12864-025-11925-y
3. Youn, S., Jeong, D., Kwon, H., **Han, E.**, Kim, S., & Jung, I. (2025). Identification of severity related mutation hotspots in SARS-CoV-2 using a density-based clustering approach. *BioData Mining*, 18, 61. doi:10.1186/s13040-025-00476-3
4. Shin, S., **Han, E.**, Seong, H., Kim, Y.I., Jung, I., & Jung, W. (2025). De novo transcriptome assembly and gene expression analysis of *Cnidium officinale* under high-temperature conditions. *BMC Genomics*, 26, 907. doi:10.1186/s12864-025-12051-5
5. Jeon, J., **Han, E.Y.**, & Jung, I. (2023). MOPA: An integrative multi-omics pathway analysis method for measuring omics activity. *PLOS ONE*, 18(3), e0278272. doi:10.1371/journal.pone.0278272

Research Experience

Graduate Researcher — Computational Biology & Bioinformatics Lab 20XX – Present
Kyungpook National University, Daegu

- Developed multi-omics integration and pathway-analysis methods for cancer subtyping using TCGA datasets (gene expression, miRNA, methylation).
- Built machine-learning models (random forest, SHAP-based interpretation) to characterize COVID-19 immune response from EHR and cytokine-profile cohorts.
- Applied density-based clustering to identify SARS-CoV-2 mutation hotspots associated with patient severity from large-scale genomic databases (GISAID).

Technical Skills

- **Programming:** [Python, R], [Bash]; *add others you use*
- **Bioinformatics:** RNA-seq / DEG analysis, de novo transcriptome assembly, pathway & GO/KEGG enrichment, multi-omics integration, single-cell RNA-seq
- **Machine learning:** clustering, random forests, SHAP / explainable AI, feature selection, benchmarking
- **Tools:** Git, [Linux/HPC], [add frameworks]

Presentations, Awards & Service

- *[Add conference talks / posters, e.g., name, venue, year.]*
- *[Add scholarships, fellowships, or awards.]*
- *[Add peer-review or academic service, if any.]*

References

Prof. Inuk Jung

Ph.D. Advisor

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Additional references available upon request.